

Matthew Mair

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EDUCATION

Virginia Tech University, Blacksburg, VA 08/24 - Present

Ph.D. student: Department of Agricultural and Applied Economics

Interfaces of Global Change Ph.D. fellow: Global Change Center

Appalachian State University, Boone, NC Degrees conferred 05/24

Bachelor of Arts: Economics (Environmental Economics and Policy)

Bachelor of Science: Political Science (International and Comparative Politics)

Chancellor's Scholarship Recipient and University Honors College Student

GPA: 4.0

East Mecklenburg High School, Charlotte, NC Degree conferred 06/20

International Baccalaureate Diploma Program Graduate

GPA: 4.71875 (Weighted), Salutatorian

PUBLICATIONS

Tapas, M.R., Howard, G., Etheridge, R., **Mair, M.**, Peralta, A. (2025) "Integrating human decision-making into a hydrological model to accurately estimate the impacts of agricultural policies." *Communications Earth & Environment* 6, 412. <https://doi.org/10.1038/s43247-025-02325-x>

Tapas, M.R., Etheridge, R., Howard, G., **Mair, M.** (2025). "Framework for Stakeholder-Driven Socio-Hydrological Modeling: Conceptual Foundations for Policy Development and Evaluation to Improve Ecosystem Health." In: Pandey, M., Umamahesh, N., Das, J., Pu, J.H. (eds) *Hydrology and Hydrologic Modelling. HYDRO 2023. Lecture Notes in Civil Engineering*, vol 410. Springer, Singapore. https://doi.org/10.1007/978-981-97-7474-6_42

WORKING PAPERS

Mair, M., and Howard, G., "Farmer Willingness to Participate in Voluntary Conservation Programs: Evidence from Eastern North Carolina"

Mair, M., Moeltner, K, and Schnitzer, M. "Conflict Among Slope Ski Tourers in the Alps: Applications for Machine Learning"

Mair, M., "Machine Learning for Migration Modeling"

Mair, M., Ortez, M., Smith, M., and Widmar, N., "Warnings from the Web: Using Online Media Data to Anticipate Crop Conditions"

RESEARCH EXPERIENCE

<i>Research Assistant, Virginia Tech University</i>	08/24 - Present
<i>Honors Thesis, Appalachian State University</i>	06/23 - 05/24
<i>Research Assistant, Appalachian State University</i>	03/22 - 03/24
<i>REU Coastal Studies Researcher, Coastal Studies Institute</i>	06/23 - 07/23
<i>Research Intern, North Carolina Institute for Climate Studies</i>	06/22 - 10/22
<i>Nutritional Assessment Coordinator, Appalachian State University</i>	03/22 - 05/22

AWARDS AND FUNDING

<i>Berkeley Sloan Grad Camp Travel Award (\$1000)</i>	08/25
<i>VT Kline Graduate Fellowship (\$907)</i>	05/25
<i>ASU Chancellor's Scholarship (full-ride merit scholarship)</i>	08/20 - 05/24
<i>ASU Chancellor's List</i>	08/20 - 05/24
<i>ASU Department of Economics Brashear Prize for Best Paper (1st Place)</i>	04/24
<i>ASU Department of Economics Top Student</i>	04/24
<i>ASU Best Undergraduate Poster (1st Place)</i>	04/24
<i>ASU College of Arts and Sciences Student and Faculty Excellence (SAFE) Grant (\$400)</i>	04/24
<i>Rocky Mountain Biological Laboratory REU Travel Award (\$1200)</i>	04/24
<i>Honors College Special Research Grant (\$800)</i>	04/24
<i>ASU Department of Government and Justice Studies Outstanding Senior Award</i>	04/23
<i>Cochlear Foundation National Inspirer</i>	11/22
<i>Barnes Program Student Research Grant (\$1200)</i>	10/22
<i>East Mecklenburg High School Salutatorian</i>	06/20

NOTABLE PRESENTATIONS

<i>Machine Learning for Migration Modeling</i>	
Virginia Tech Nutshell Games (Science Public Communication)	10/25
University of California Berkeley & Sloan Grad Camp	08/25
<i>Documenting Shoreline Loss in Virginia Using Satellite Imagery</i>	
Virginia Tech Office of Geographic Information Systems 2025 Symposium	04/25
<i>Factors that Impact Farmer Willingness to Participate in Conservation Programs</i>	
Appalachian State University Honors College Thesis Defense	04/24
Appalachian State University OSR Student Research Day	04/24
Association for Environmental Studies and Sciences Student Research Symposium	04/24
National Conference for Undergraduate Research	04/24
Elon University Environmental Policy Workshop	02/24
State of North Carolina Undergraduate Research and Creativity Symposium	11/23

North Carolina Honors Association Annual Conference	11/23
Resilience and Adaptation to Coastal Change REU Symposium	11/23
East Carolina University Summer Undergraduate Research Symposium	07/23

<i>International Climate Negotiations: My Experience at COP27</i>	
Appalachian State University Global Symposium	11/23

<i>The Effect of Voting for Redistribution on Cooperation</i>	
Appalachian State University Annual Dean's Club Research Symposium	10/23
Appalachian State University Dean's Club Research Series	03/23

<i>Understanding Sea Level Rise and Its Impacts in North Carolina</i>	
Southern Regional Honors Conference	03/23
North Carolina Coastal Resources Commission Science Panel Meeting	11/22

INTERNATIONAL EXPERIENCE

<i>Student</i> , Ireland Study Abroad Experience	01/23
<i>Observer</i> , United Nations Climate Change Conference, Sharm El-Sheikh, Egypt	11/22

LEADERSHIP EXPERIENCE

<i>President and Founder</i> , ASU Honors Vanguard Leadership Board	04/23 - 05/24
<i>Treasurer and Founding Member</i> , ASU Chancellor's Scholars Board	10/22 - 05/23
<i>Vice President</i> , ASU Politics Club	04/22 - 05/23
<i>Interim Director of the Treasury</i> , ASU Student Government Association	03/22 - 10/22
<i>Presidential Campaign Manager</i> , ASU Student Government Association	10/21 - 02/22
<i>Senator</i> , ASU Student Government Association	10/20 - 05/21

SERVICE AND MENTORSHIP

<i>Honors Peer Mentor</i> , ASU Honors College	03/21 - 05/24
<i>Honors Vanguard Ambassador</i> , ASU Honors College	08/20 - 05/24

PROFESSIONAL DEVELOPMENT

<i>Grad Camp for Environmental and Energy Economics</i> , UC Berkeley - Berkeley, CA	08/25
<i>Introduction to Marine Energy</i> , Atlantic Marine Energy Center - Durham, NH	08/25
<i>Economic Forces in American History</i> , Foundation for Teaching Economics - Charleston, SC	07/24
<i>Seminar on Classical Liberalism</i> , AIER - Crawfordsville, IN	06/24
<i>Valuing Nonmarket Benefits</i> , Society for Benefit-Cost Analysis - Online	07/23

TOOLS

R, STATA, Google Earth Engine, LaTeX

SKILLS

Machine Learning, Decision Trees, Discrete Choice Analysis, Environmental Valuation, Policy Analysis